

Cybersecurity

Splunk Searches

SIEM Day 2



Learning Objectives

By the end of today's lesson, you will be able to:

- 1 Explore and select Splunk add-ons and apps based on project needs.
- 2 Upload logs into a Splunk repository.
- 3 Write complex Splunk Processing Language (SPL) queries to analyze specific security situations.



Let's Recap



Recap

Before we introduce Splunk and its capabilities, let's review the concepts covered in the last lesson:

- 1 Organizations use **continuous monitoring** to monitor risks to the confidentiality, integrity, and availability of their technical assets.
- 2 Organizations use **logs** that contain **log entries** to monitor against these risks.
- 3 Organizations **aggregate, parse, and normalize** multiple logs so that they can be analyzed together.
- 4 Organizations correlate these logs with **correlation rules** to alert when a security event or suspicious activity is detected.
- 5 **SIEM** software is a security tool that can assist with all of the above processes.

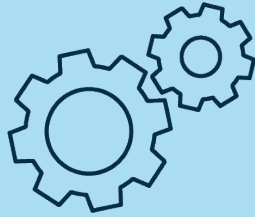
We also learned about the many SIEM vendors available, each with different features, strengths, and weaknesses.



In the Next Two Modules...

... we will:

- 1 Focus on one of the most popular SIEM vendors: Splunk.
- 2 Learn about Splunk and its features.
- 3 Complete hands-on activities within Splunk that mirror the actions that security professionals perform every day.



Splunk **Capabilities**



Splunk is the vendor name of a **big data software solution**, and the SIEM tool is just one of the thousands of features Splunk provides.

Splunk Capabilities

Splunk is...

A software tool that searches, analyzes, and monitors big data with an easy-to-use interface.

Splunk can...

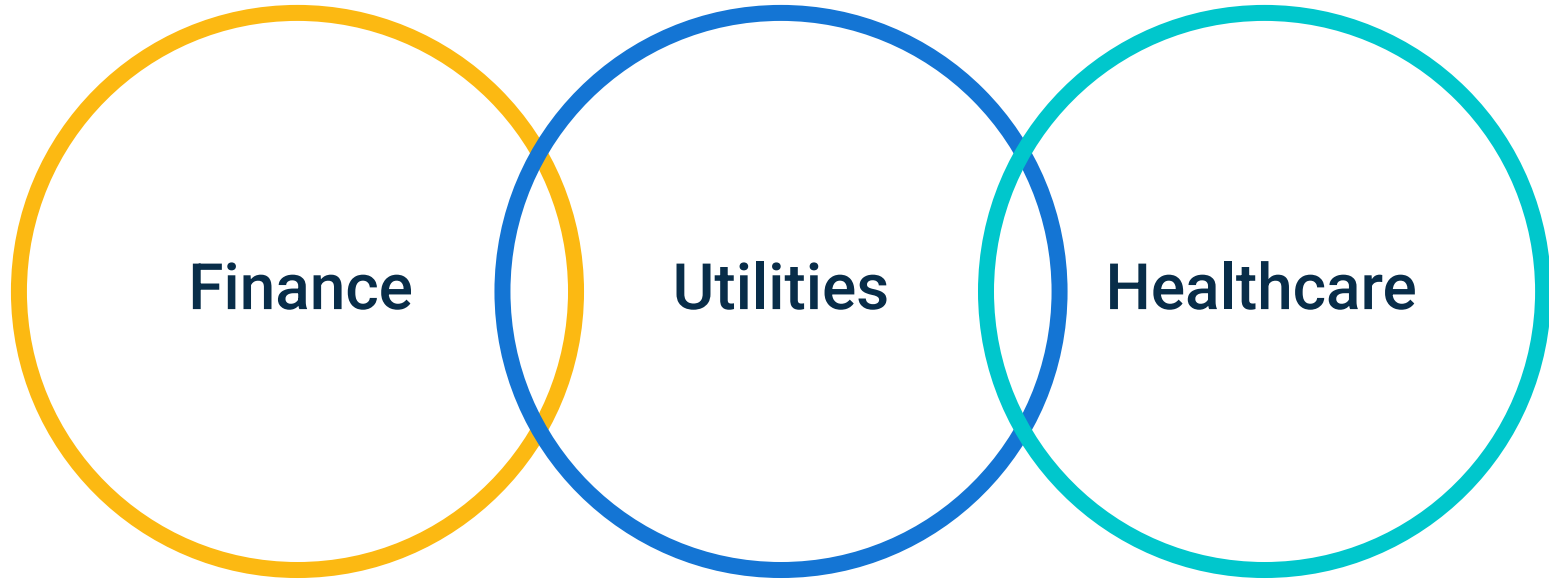
Capture large amounts of incoming data, which can be used to create visualizations, reports, and alerts.

Splunk has...

A base product that is designed to conduct basic tasks such as searching and reporting.

Splunk Capabilities

This week, we're focusing on Splunk's benefits to the InfoSec industry. **However**, Splunk is useful for a variety of industries, such as:



Finance

Financial organizations can use Splunk to analyze mortgage rates and determine future rate changes.



Utilities

Gas companies can use Splunk to monitor customer use levels.

Healthcare

Medical researchers can use Splunk to create reports and metrics for analyzing the success of medical trials.



Apps, Add-Ons, and Suites

Splunk can be used for these industry-specific tasks by adding the following to the base product: **Splunk apps**, **Splunk add-ons**, and **Splunk suites**.



Splunk Apps

These are applications that users can add to their Splunk base product that have custom searches and features, and their own interfaces.

App Type: App ✕

Showing 1-20 of 1079 results

Newest ▾

 Predictive Crime Showcase 15 Installs	 Perseus - An Analyst-Friendly IR 15 Installs	 Metricator application for 115 Installs	 People and Vehicle Analytics App for 2 Installs
 Covid19 85 Installs	 Splunk Connect for Mission Control 17 Installs	 BlueCat DNS Edge for Splunk 26 Installs	 Trend Micro Email Security for Splunk 3 Installs
 Deep Learning Toolkit for Splunk 13 Installs	 Scalable Vector Graphics - Custom 533 Installs	 Sandfly Security 0 Installs	 AWS Trusted Advisor Aggregator 122 Installs

Splunk Add-Ons

These are smaller Splunk components that provide additional functionality without their own interfaces.

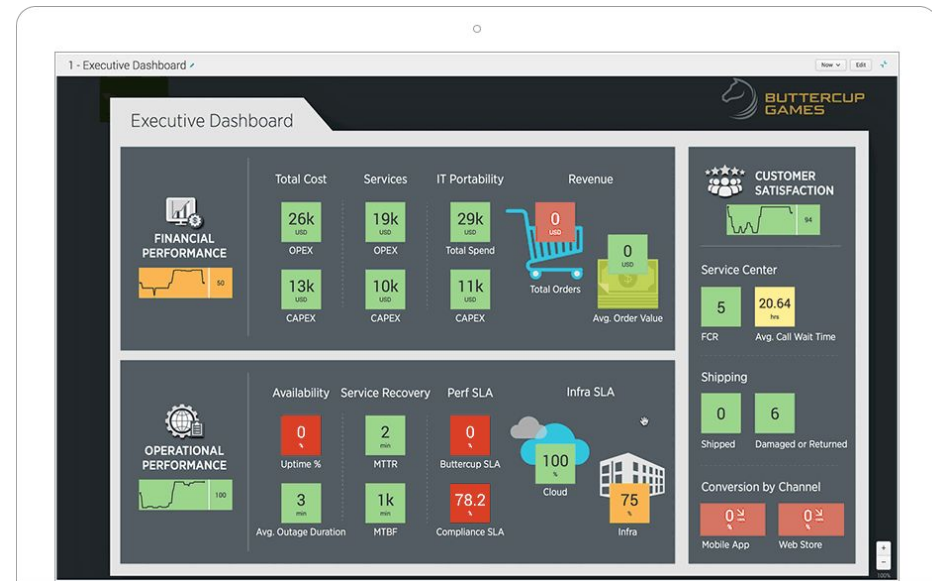
App Type: Add-on ✕

Showing 1-20 of 867 results Newest ▾ ☰ ☒

 Radware Cloud DDoS Add-On 6 Installs	 ExtraHop Add-On for Splunk 78 Installs	 BlueCat DNS Edge Technical Add-on for Splunk 57 Installs	 Trigger LogicHub Stream 0 Installs ✓
 Sandfly Security Add-on for Splunk 2 Installs	 Sixgill Darkfeed 2 Installs ✓	 Cisco WebEx Meetings Add-on for Splunk 37 Installs ✓	 API Fortress - Splunk Connector Hosted Externally
 RocketChat Alert Action Hosted Externally	 Splunk ODBC 0 Installs 🔧	 TA for finnhub.io - Stock data 4 Installs	 Technology Add-On for Vectra Cognito 175 Installs ✓

Splunk Suites

These are collections of Splunk apps with a single focus, such as an industry or technology.



We will not review **Splunk suites** in this lesson.



Splunk IT Service Intelligence (ITSI)

Simplify operations, prioritize problem resolution and align IT with the business using a monitoring and analytics solution tailored for today's environments.

[Get Predictive Analytics >](#)



VictorOps

Empower your on-call teams to find and fix problems faster with automated and insightful incident response routing, collaboration and reviews.

[Make On-Call Suck Less >](#)



Splunk Insights for AWS Cloud Monitoring

Don't lose sight or control of your data. Enjoy end-to-end security, operational and cost-management insights for your AWS workloads.

[See Through the Cloud >](#)



Splunk App for Infrastructure

Unify and correlate logs and metrics on one solution. Get free comprehensive infrastructure monitoring, alerting and investigation with your Splunk Enterprise license.

[Install to Insights in Minutes >](#)

Apps, Add-Ons, and Suites

Splunk has so many of these apps and add-ons that they are broken up by type.

Technology

For example, there are apps and add-ons specific to cloud servers.

Vendor

For example, there are apps and add-ons specific to the security vendor Rapid7.

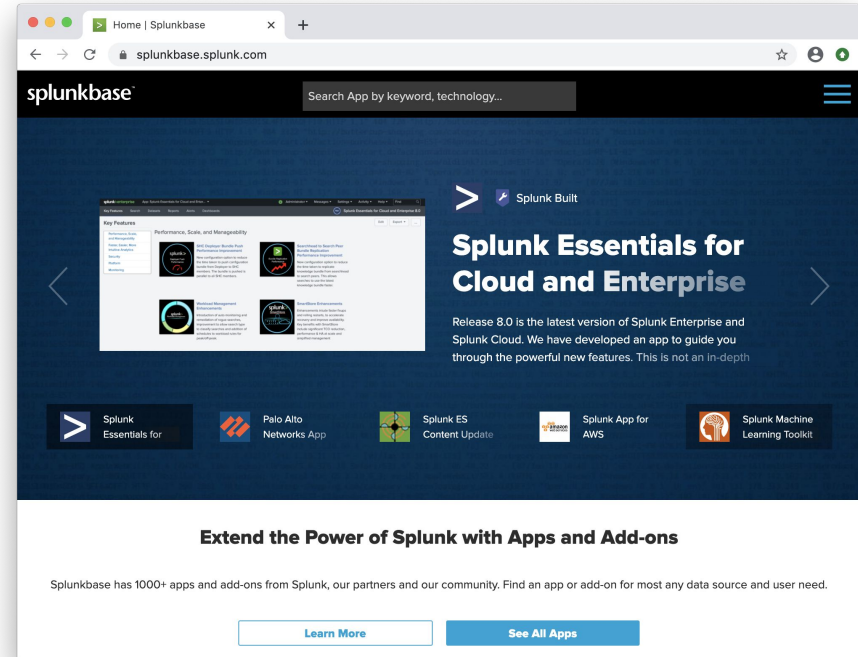
Industry

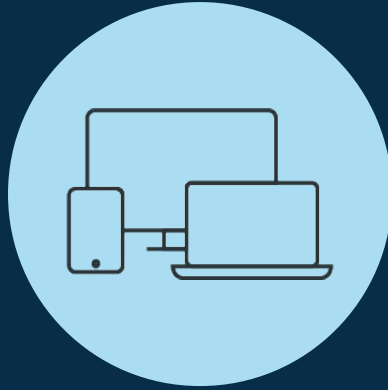
For example, there are apps and add-ons specific to manufacturing organizations.

Splunk Apps and Add-Ons

We will explore various Splunk apps and add-ons with the following scenario:

- Your manager has notified you that the organization has purchased a web application filter from the vendor F5.
- Your manager would like you to find the appropriate Splunk app to assist with monitoring this product.





Instructor Demonstration

Splunk apps and add-ons

Splunk Account

Do not sign up for a Splunk account at this time.

There is a 7-day trial window that we will start next week so that we have access to Splunk for as long as the curriculum requires it.



Create Your Splunk Account

Already Have a Splunk Account? [Login](#)

First Name

Last Name

Email Address

Phone Number

Company

United States ▼

Postal Code

Password

☐ I agree to the [Splunk General Terms](#) and [Splunk Website Terms & Conditions of Use](#).

Create Your Account



Questions?





Activity

Splunk features

In this activity, you will analyze Splunk apps and add-ons to determine which will work with your security products.

Suggested Time
15 minutes



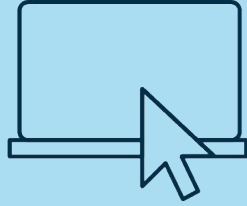


Time's Up!
Let's review



Questions?

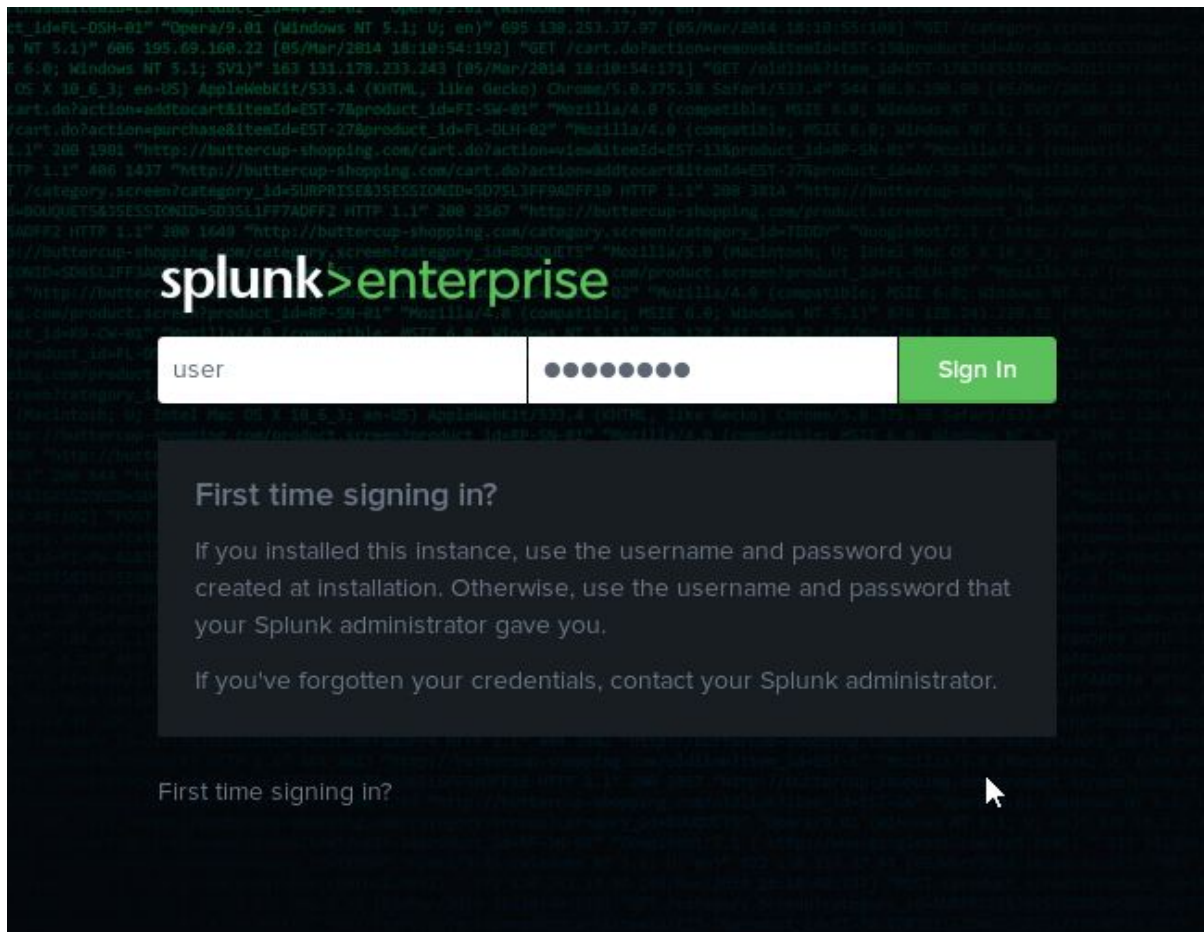




Tour of Splunk

Tour of Splunk

The best way to learn how to use the Splunk product is to dive right in and start using the application.



First time signing in?

If you installed this instance, use the username and password you created at installation. Otherwise, use the username and password that your Splunk administrator gave you.

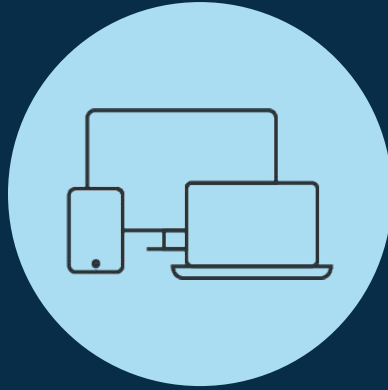
If you've forgotten your credentials, contact your Splunk administrator.

First time signing in?

Tour of Splunk

In the following walk-through, we'll explore the Splunk:





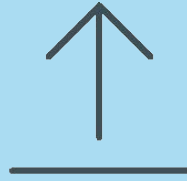
Instructor Demonstration

Splunk tour



Questions?





Adding Data into Splunk



Before we add data,
it is important to
understand **Splunk's**
architecture and how it
handles incoming data.

Splunk Architecture Basics

Splunk architecture contains two primary components:

01

The Indexer

- Splunk transforms incoming data into **events**.
- Splunk adds these events into repositories called **indexes**.
- **Indexers** are used to add events to indexes and search through the data.

02

The Search Head

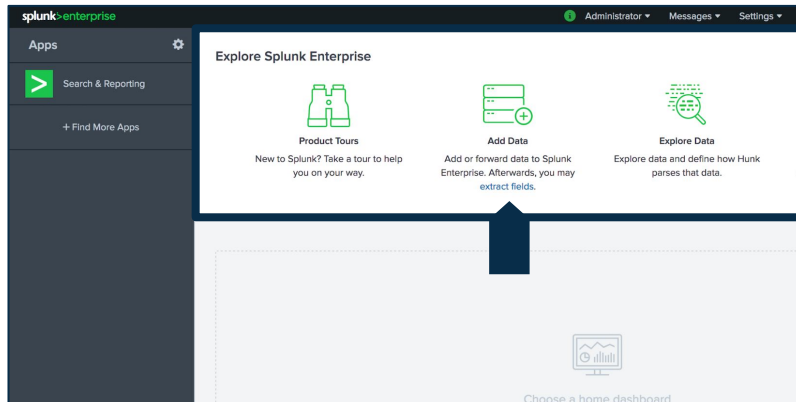
- The **search head** is Splunk's GUI that we use to conduct searches.
- It manages search requests to the indexer and provides the search results back to the user.

Splunk Data Addition Methods

To add data to Splunk on the **Add Data** page, we use one of the following paths:

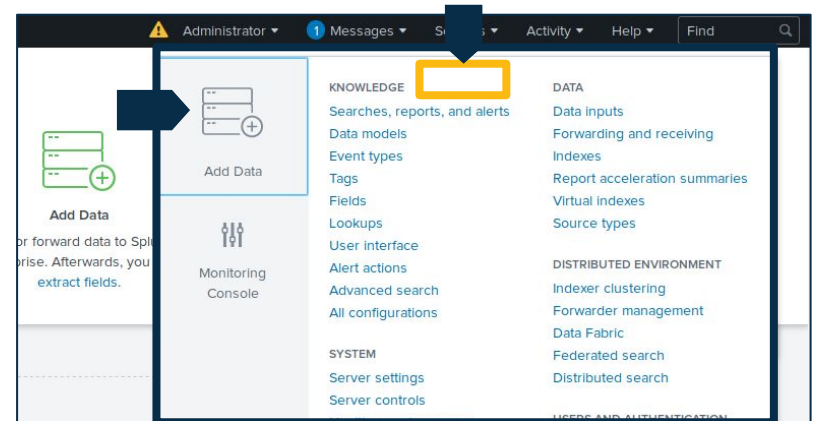
From the Welcome Page

Go to **“Explore Splunk Enterprise”**
and select **“Add Data.”**



From the Search & Reporting App

Select **“Settings”** and then
“Add Data.”



Splunk Data Addition Methods

The Add Data page prompts you to add data either by:

- Data source
- Specific method

Follow guides for onboarding popular data sources

Cloud computing

Get your cloud computing data in to the Splunk platform.

10 data sources

Networking

Get your networking data in to the Splunk platform.

2 data sources

OS Operating System

Get your operating system data in to the Splunk platform.

1 data source

Security

Get your security data in to the Splunk platform.

3 data sources

4 data sources in total

Or get data in with the following methods

Upload

files from my computer

Local log files
Local structured files (e.g. CSV)
[Tutorial for adding data](#)

Monitor

files and ports on this Splunk platform instance

Files - HTTP - WMI - TCP/UDP - Scripts
Modular inputs for external data sources

Forward

data from a Splunk forwarder

Files - TCP/UDP - Scripts

Adding Data: Data Source

Adding data by data source lets us upload various types of data.

- 1 For example, a Splunk user may want to add Palo Alto Firewall logs into Splunk.
- 2 The Palo Alto option under Networking is an example of a data type.
- 3 Based on the option selected, an add-on may be provided or settings configured.

Adding Data: Specific Method

This feature let you add data by one of the following methods:

Monitor

Splunk monitors logs from a system, device, or application to which it has direct access.

This method is commonly used by businesses to monitor their production environment.

Forward

Install a program called a forwarder on the system from which logs are collected.

Forwarders forward logs from a device into the Splunk system.

Upload

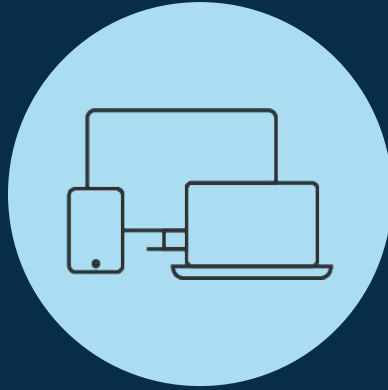
Manually upload logs directly into your Splunk repository.

While monitoring and forwarding are important to understand conceptually, we will primarily use the upload process for the rest of this lesson.

Uploading Data into Splunk

In this walk-through, we will use the following scenario to upload data into Splunk:

- 1 Your manager has reported some suspicious login activity on your Linux servers.
- 2 They have provided you with the login activity.
- 3 You must upload this activity data into Splunk so that it can be analyzed.



Instructor Demonstration

Data upload



Activity

Uploading data into Splunk

In this activity, you will upload several log files that will be used later to analyze security events.

Suggested Time
15 minutes





Time's Up!
Let's review



Questions?

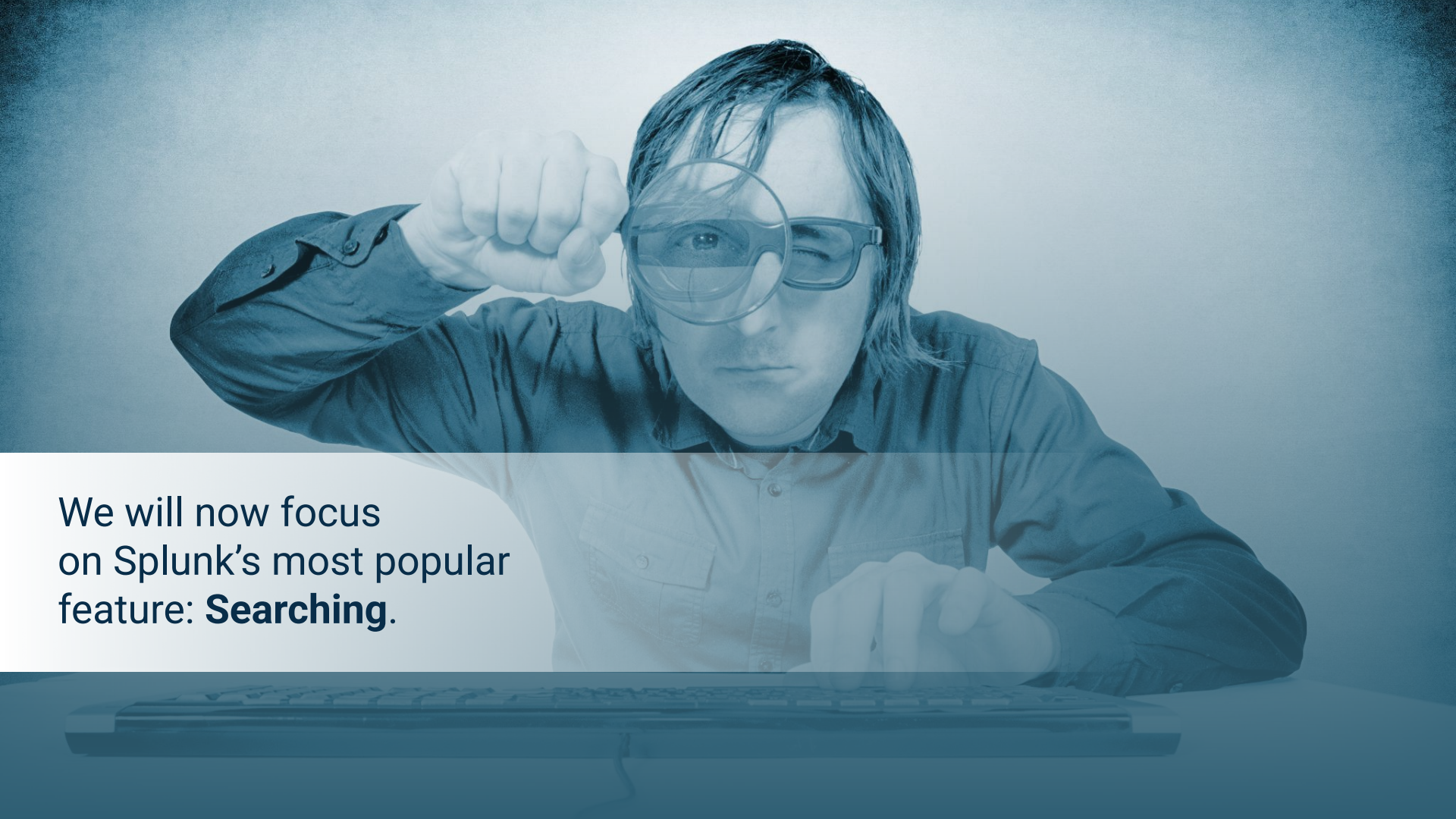




Break
15 mins



Searching with **Splunk**



We will now focus
on Splunk's most popular
feature: **Searching.**

Searching with Splunk

Searching in Splunk allows users to query uploaded and monitored data.

The screenshot shows the Splunk Search & Reporting interface. At the top, the navigation bar includes 'splunk>cloud', 'App: Search & Reporting', 'Messages', 'Settings', 'Activity', and a 'Find' search bar. Below this, the 'Search & Reporting' section is active, displaying a 'New Search' window. The search query is 'categoryid=sports'. The results show 115 events from 4/6/21 6:04:59.000 PM to 4/7/21 6:04:56.000 PM. A bar chart visualization is shown above a table of events. The table has columns for 'Time' and 'Event'. The first event is a POST request to /cart.do?action=purchase&itemId=EST-21&JSESSIONID=SD0SL9FF7ADFF52798 H TTP 1.1" 200 2383 "http://www.buttercupgames.com/cart.do?action=addtocart&itemId=EST-21&categoryId=SPORTS&productId=C U-PG-G06" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_7_4) AppleWebKit/536.5 (KHTML, like Gecko) Chrome/19.0.1084.46 S afari/536.5" 527. The second event is a POST request to /product.screen?productId=CU-PG-G06&JSESSIONID=SD0SL9FF7ADFF52798 HTTP 1.1" 200 3884 "http://www.buttercupgames.com/category.screen?categoryId=SPORTS" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_7_4) AppleWebKit/536.5 (KHTML, like Gecko) Chrome/19.0.1084.46 Safari/536.5" 986. Both events have a host of 'www2' and a source of 'tutorialdata.zip:www2/access.log'.

splunk>cloud App: Search & Reporting Messages Settings Activity Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search Save As Create Table View Close

"categoryid=sports" Date time range

✓ 115 events (4/6/21 6:04:59.000 PM to 4/7/21 6:04:56.000 PM) No Event Sampling Job standard_perf (search default) Smart Mode

Events (115) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 hour per column

List Format 20 Per Page Prev 1 2 3 4 5 6 Next

< Hide Fields All Fields

SELECTED FIELDS

- a host 3
- a source 3
- a sourcetype 1

INTERESTING FIELDS

- a action 5
- # bytes 100+
- a categoryid 1
- a clientip 66

i	Time	Event
>	4/7/21 5:12:50.000 PM	201.42.223.29 - - [07/Apr/2021:17:12:50] "POST /cart.do?action=purchase&itemId=EST-21&JSESSIONID=SD0SL9FF7ADFF52798 H TTP 1.1" 200 2383 "http://www.buttercupgames.com/cart.do?action=addtocart&itemId=EST-21&categoryId=SPORTS&productId=C U-PG-G06" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_7_4) AppleWebKit/536.5 (KHTML, like Gecko) Chrome/19.0.1084.46 S afari/536.5" 527 host = www2 source = tutorialdata.zip:www2/access.log sourcetype = access_combined_wcookie
>	4/7/21 5:12:48.000 PM	201.42.223.29 - - [07/Apr/2021:17:12:48] "POST /product.screen?productId=CU-PG-G06&JSESSIONID=SD0SL9FF7ADFF52798 HTTP 1.1" 200 3884 "http://www.buttercupgames.com/category.screen?categoryId=SPORTS" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_7_4) AppleWebKit/536.5 (KHTML, like Gecko) Chrome/19.0.1084.46 Safari/536.5" 986 host = www2 source = tutorialdata.zip:www2/access.log sourcetype = access_combined_wcookie

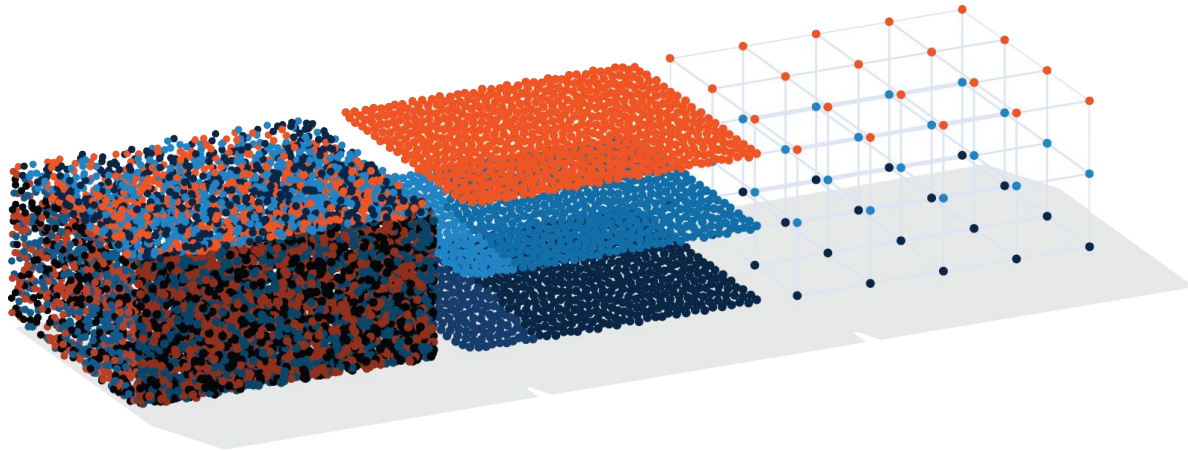
Splunk queries
can be customized
to search only for
specific data or to
manipulate how the
data is displayed.

Searching in Splunk

We can use Splunk queries to find specific, helpful information about a security event.

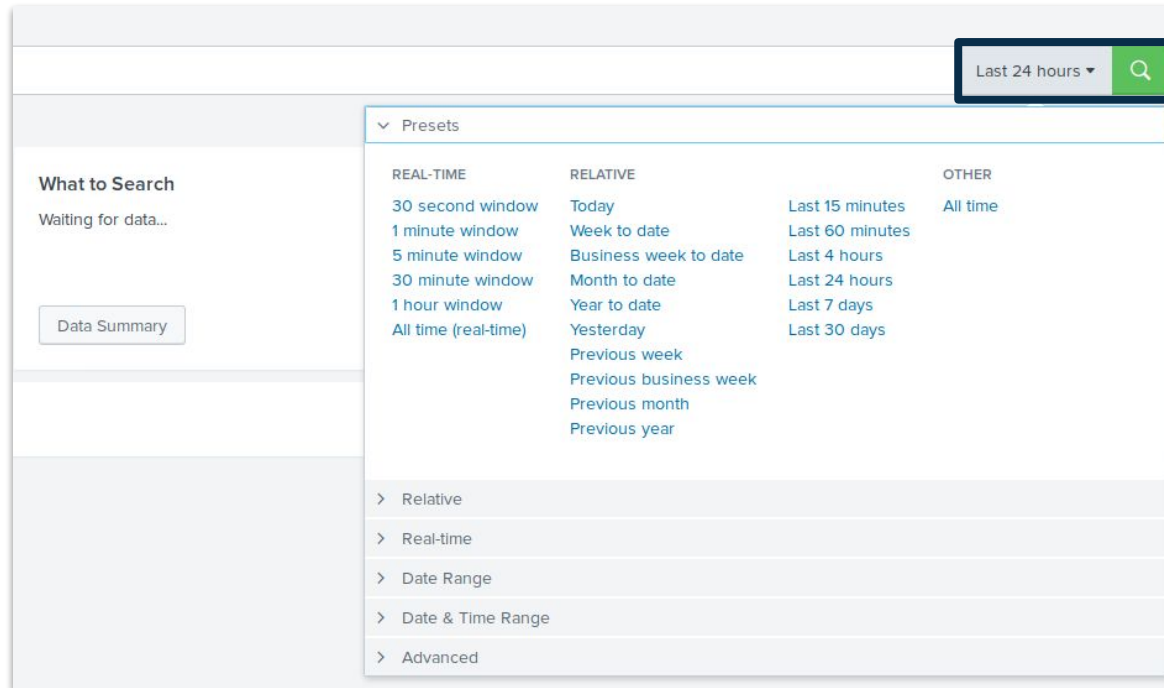
For example, we can use Splunk to:

- Determine the **primary IP** that is being attacked during a DDoS attack.
- Determine the **user ID** that is being used in a brute-force attack.



Searching in Splunk

Splunk searching is almost always a time-based search.



All events have associated timestamps.

To search for events, we must designate a time range or real-time period.

Searching in Splunk

A user can select the following:

Real-Time Search

Returns a window of real-time data as it is happening and continues to update as the events occur.

Relative Search

Returns data by date, date range, time, or time range. Results will not change even if more events occur.

All-Time Search

Returns all available data based on the search.

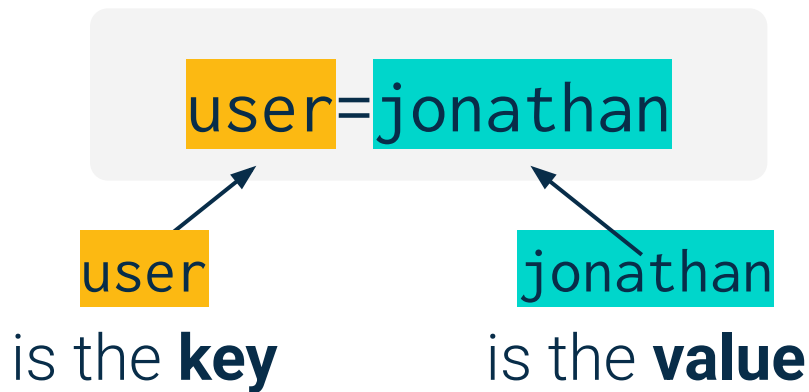


Splunk queries are designed using a coding language called **Splunk processing language (SPL)**.

Key-Value Pairs

Using key-value pairs is the most common way to search for data by matching keywords with specific information (values).

For example: If you want to find a user named **jonathan** in your search results, you would design the following search:



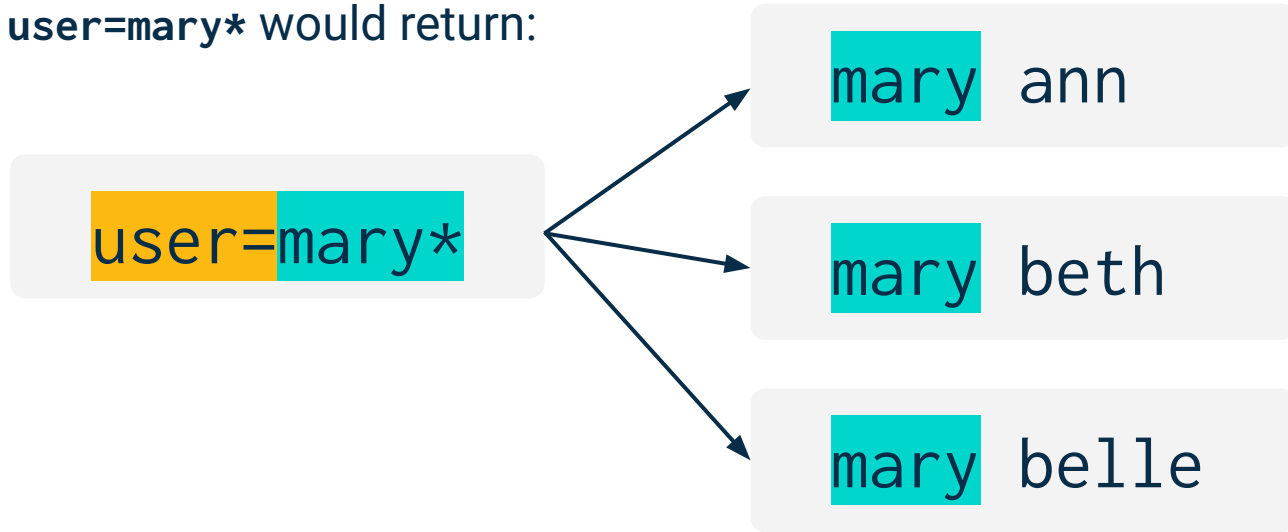
`user=jonathan`
is the **key-value pair**

Wildcards

Similar to other programming languages, SPL uses **wildcards**. When used with the wildcard symbol (*), the search results return the search term followed by any character or string in place of the wildcard symbol.

For example:

`user=mary*` would return:

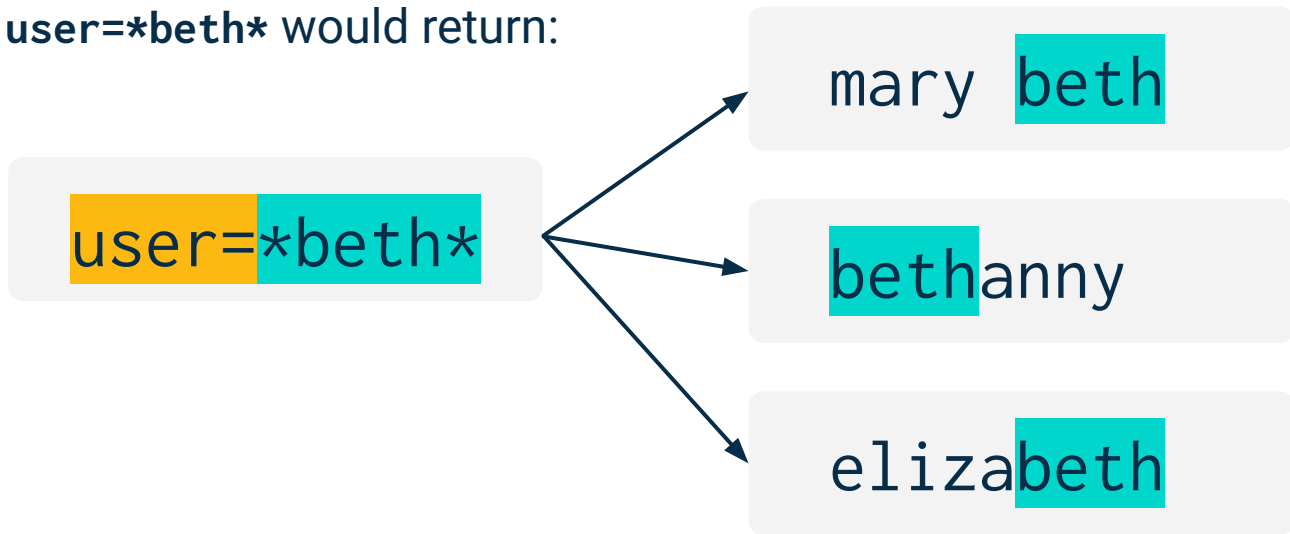


Wildcards

Wildcards can also be used to find a value surrounded by any character.

For example:

`user=*beth*` would return:



Boolean Expressions

SPL uses the Boolean expressions **AND**, **OR**, and **NOT** to assist in searching for specific data.

Expression	Use	Example
AND	Combines two key-value searches	user=jonathan AND activity=login
OR	Looks for multiple instances of a key-value pair	user=jonathan OR user=beth
NOT	Excludes certain values from search results	user=jonathan NOT activity=logout

Search Demonstration

We will use the following scenario:

01

Your manager has reported some suspicious login activity on your Linux servers.

02

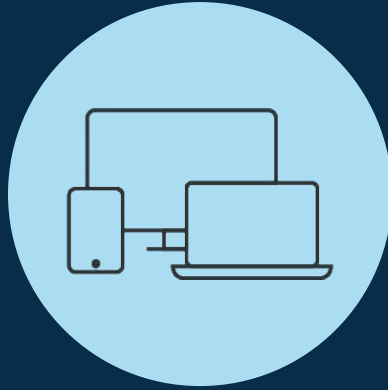
They would like you to write a query to look at these activities, specifically for logins coming from the source **IP 10.11.36.17**.

03

They believe this IP is from a machine infected with malware.



Note: **src_ip** is the field name for the source IP.



Instructor Demonstration

Searching



Activity

SPL search

In this activity, you will design SPL searches to run against the vulnerability-scanning log file `nessus.txt`.

Suggested Time
15 minutes





Time's Up!
Let's review



Questions?





Searching Fields with **Splunk**

Searching Fields with Splunk

So far we've manually typed out the keys and values for our SPL queries.

The more complex the queries become, the more time-consuming this task will be.

Sometimes we don't know the values or the format of the values that exist in the search results.





Each server and application creates their own **key** and **value names**.

Complexities of SPL Queries

For example: If we need to find users that logged in to a machine:

The **key** might be:

Activity

Event_type

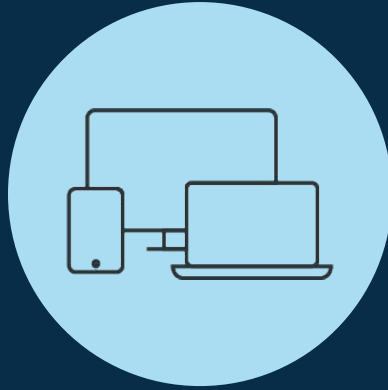
User_activity

The **value** might be:

Login

Logon

Logged In



Instructor Demonstration

Splunk fields

Search Fields

When files are uploaded and parsed, the data is separated into fields, as shown on the left side of Splunk's search page.

< Hide Fields	
SELECTED FIELDS	
a host	1
a source	1
a sourcetype	1
INTERESTING FIELDS	
a action	1
a app	1
a date_hour	1
a date_mday	1
a date_minute	9
a date_month	1
a date_second	50
a date_wday	1
a date_year	1
a date_zone	1
a dest	1

Default Fields

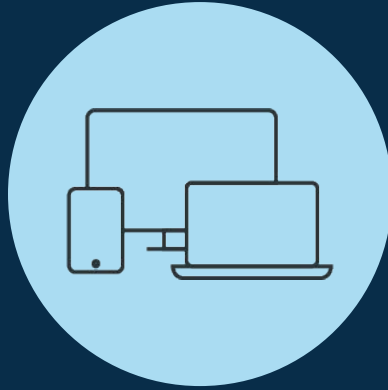
Appear in every log event.

Interesting Fields

Appear in at least 20% of the log events.

Value Count

A number on the right of each field indicating the count of different values for that field.



Instructor Demonstration

Creating queries by selecting fields



Activity

Searching fields with Splunk

In this activity, you will create complex SPL queries by selecting fields in your Splunk search.

Suggested Time

15 minutes



Time's Up!
Let's review



Questions?





Advanced Searches with **Piping**

SPL Piping

We can add piping to our SPL queries to modify or adjust the display of the results, or to create custom reports.

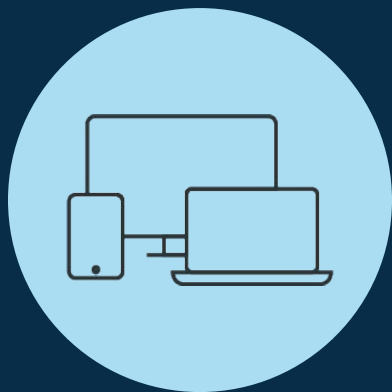
```
source="Linux_login.csv" host="Linux_Server__" sourcetype="csv" | head 20 | sort src_ip
```

SPL piping uses the | symbol in the search queries.



Piping works in Splunk as it does in Linux:

the data is modified from left to right as it flows through the pipeline.



Instructor Demonstration

SPL piping



Activity

Advanced searches with piping

In this activity, you will run several advanced searches to find out whether a specific user is being targeted by an attacker.

Suggested Time
10 minutes





Time's Up!
Let's review



Questions?





Thank You